



NVMe SSDs

Gen 5 is here

In the rapidly evolving world of NVMe SSDs, the past year has been a testament to remarkable growth and innovation. Despite an impressive 80% growth in the internal SSD segment in India, the number of new launches has been modest. This growth is primarily attributed to the shift from hard drives to SSDs in laptops. The Indian market is predominantly occupied by PCIe Gen 3 and Gen 4 NVMe SSDs, while the NVMe Gen 5 SSDs, known for their high cost and tendency to run hot, are slowly making their presence felt. These Gen 5 SSDs often come equipped with exotic heatsinks featuring heat pipes, catering to their cooling needs. Just like the past year and the year before that, Phison continues to dominate the third-party SSD controller market. All the top designs are currently powered by Phison controllers. The mainstream brands haven't launched that many SSDs this year. It may take some time before PCIe Gen 5 becomes mainstream, considering the heat and cost factors. This was the case back when Gen 4 SSDs started appearing. Heatsinks became more of a necessity to ensure sustained performance otherwise there was a clear and marked performance degradation. Interestingly, the price gap between SSDs and hard drives is narrowing, so we're looking at a near future where even budget laptops and PCs will predominantly feature SSDs over HDDs.

Zero1 Award Winner: AORUS Gen 5 10000 SSD

The AORUS Gen 5 10000 SSD, crowned as this year's winner, stands out with its Phison E26 controller. It boasts an optional heatsink, essential for optimal performance. This SSD impresses with sequential read speeds reaching approximately 10,000 MBps, not yet the peak of PCIe Gen 5's capability, as evidenced by their faster Gen 5 12000 SSD which has been making some waves in international markets. Latency-wise, the unit shows a similar performance curve as the Seagate FireCuda 530, which was a flagship from Seagate among Gen 4 SSDs. So you can say that the



AORUS Gen 5 10000 SSD
Price: ₹29,572

AORUS Gen 5 10000 is better in raw performance without taking a hit in the latency scores. The unit comes with a very tall heatsink with not one but two heat pipes. It does get a little hot but as we mentioned previously, this is typical of first-generation launches. While power efficiency is compromised, typical for a first-generation SSD, its high performance more than compensates, leading to its top spot in this year's Zero1 Awards.

Runner-Up: Kingston FURY Renegade

The Kingston FURY Renegade, our runner-up, is a mature product based on the Phison E18 controller, a staple for PCIe Gen 4 SSDs. It delivers sequential read and write speeds of 7300 MBps and 6000 MBps, respectively, and comes with a heatsink. It doesn't take full advantage of the E18 platform since we have seen designs with write speeds hitting 7000 MBps in the past. While its performance could be better, the balance it strikes between performance and price-per-GB makes it a strong contender in the market.



Kingston FURY Renegade
Price: ₹18,000

DIGIT BEST BUY WINNER-2023



WD Blue SN580
Price: ₹10,819 (2 TB)

Best Buy: WD Blue SN580

For those seeking value for money, the WD Blue SN580 emerges as the 'Best Buy'. Its competitive pricing, paralleling hard drives, combined with performance that is nearly 30 times greater, offers exceptional value. The cost-per-GB of the SN580 comes down to Rs.53/GB which is quite great compared to its peers which sport a slight price premium for the performance that they boast. It exhibits sequential read and write speeds of about 4150 MBps, so you've got performance and great value for money in this package.

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